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Immunogenicity and Safety of a Hexavalent DTwP-IPV-HB-PRP~T Vaccine Versus Separate DTwP-HB-PRP~T, bOPV, and IPV Vaccines Administered at 2, 4, 6 Months of Age Concomitantly With Rotavirus and Pneumococcal Conjugate Vaccines in Healthy Infants in Thailand

Leilani Sanchez, MD, MSc, Supattra Rungmaitree, MD,† Pope Kosalaraksa, MD,‡ Watsamon Jantarabenjakul, MD, PhD,§¶ Julie Leclercq, MSc,‖ Yuvadee Yaiprayoon, MSc,** Venkata Jayanth Midde, MB, BS,†† Kucku Varghese, PhD,‡‡ Somnath Mangarule, MD,§§ and Fernando Noriega, MD‡‡*

Background: This study investigated the immunogenicity and safety of a fully liquid, hexavalent, diphtheria (D)-tetanus (T)-whole-cell pertussis (wP)-inactivated poliovirus (IPV)-hepatitis B (HB)-*Haemophilus influenzae* b (PRP-T) vaccine compared to licensed DTwP-HB-PRP~T, IPV, and bivalent oral poliovirus (bOPV) vaccines following co-administration with other pediatric vaccines [pneumococcal conjugate vaccine (PCV13) and rotavirus vaccine].

Methods: Phase III, randomized, open-label study in Thailand. Healthy infants received DTwP-IPV-HB-PRP~T at 2, 4 and 6 months of age (N = 228), or DTwP-HB-PRP~T and bOPV (2, 4 and 6 months of age) and IPV (4 months of age) (N = 231). All participants received PCV13 (2, 4 and 6 months of age) and rotavirus vaccine (2 and 4 months of age). Immunogenicity for all antigens was assessed using validated assays, and noninferiority post-third dose was evaluated for anti-D, anti-T, anti-pertussis [anti-pertussis toxin (anti-PT) and anti-fimbriae 2/3 (anti-FIM)], anti-polio 1, 2, 3, anti-HB, and anti-PRP~T. Safety was assessed using parental reports.

Results: Noninferiority was demonstrated for each antigen, and overall noninferiority of DTwP-IPV-HB-PRP~T versus DTwP-HB-PRP~T+bOPV+IPV was concluded. Similarity in each group was observed for the GMC ratio for antirotavirus antibodies (20.9 and 17.3, respectively) and anti-PCV13 antibodies (range: 8.46–32.6 and 7.53–33.1, respectively). Two serious adverse events were related to DTwP-IPV-HB-PRP~T (febrile convulsion and acute febrile illness) and 1 was related to DTwP-HB-PRP~T+bOPV+IPV (febrile seizure), but overall there were no safety concerns with similar rates of participants experiencing solicited (99.1% and 98.3%) and unsolicited (19.3% and 19.5%) adverse events in each group.

Conclusions: This study confirmed the suitability of DTwP-IPV-HB-PRP~T primary series vaccination in combination with rotavirus and PCV13 vaccines.

Key Words: coadministration, hexavalent, primary, vaccine, whole-cell

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The widespread use of pediatric combination vaccines with multiple antigens in a single vaccination is important to achieve and maintain a low incidence of childhood diseases, including diphtheria (D), tetanus (T), pertussis, hepatitis B (HB), *Haemophilus influenzae* type b (Hib) infection and poliomyelitis.[1,2] Such multivalent vaccines simplify compliance with crowded pediatric vaccination schedules, enabling high vaccine coverage rates. A hexavalent vaccine containing D, T, whole-cell pertussis (wP), inactivated poliovirus (IPV), HB and Hib [purified capsular polysaccharide conjugated to tetanus toxoid (PRP~T)] (SHAN6) combines the IPV antigen from SHANIPV with the D, T, wP, HB and Hib antigens from SHAN5. These antigens are well-established, and SHANIPV and SHAN5 received marketing authorizations in India in June 2015 and March 2014, respectively. Both vaccines were subsequently prequalified by the World Health Organization (WHO). Previous studies have shown good vaccine safety, strong immunogenicity and lot-to-lot consistency for the DTwP-IPV-HB-PRP~T vaccine following primary vaccination at 6–8, 10–12 and 10–14 weeks, and a booster at 12–24 months of age.[3–5] A full list of abbreviations used within this article can be found in Table, Supplemental Digital Content 1, http://links.lww.com/INF/F71.

The present study was conducted in Thailand to provide additional immunogenicity and safety data for DTwP-IPV-HB-PRP~T to generate safety and immunogenicity data for the 2, 4 and 6-month primary series schedule, which is used in a wide range of countries, compared with licensed, antigen-matching

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From the *Sanofi, Taguig City, Philippines; †Department of Pediatrics, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok, Thailand; ‡Department of Pediatrics, Faculty of Medicine, Khon Kaen University, Thailand; §Department of Pediatrics, King Chulalongkorn Memorial Hospital, Bangkok, Thailand; ¶Department of Pediatrics, Center of Excellence for Pediatric Infectious Diseases and Vaccines, Faculty of Medicine, Chulalongkorn University, Bangkok, Thailand; ‖AIXIAL, Boulogne-Billancourt, France; **Sanofi, Thailand; ††Sanofi Healthcare India Private Limited, India; ‡‡Sanofi, Swiftwater, Pennsylvania, USA; and §§Sanofi, Marcy l'Etoile, France.

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Address for correspondence: Leilani Sanchez, MD, MSc, Global Clinical Development Strategy Expert, Vaccines, Sanofi, 21/F One World Place, 32nd St. Bonifacio Global City, Taguig City, Philippines 1634. E-mail: Leilani.Sanchez@sanofi.com.

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SHAN5 (DTwP-HB-PRP~T), IMOVAX Polio (IPV) and bivalent oral poliovirus vaccine (bOPV). Additionally, the inclusion of the pneumococcal conjugate vaccine (PCV13) and rotavirus vaccine in the Thai pediatric vaccination schedule allowed us to evaluate their co-administration in terms of safety and immunogenicity.

MATERIALS AND METHODS

Study Design and Participants

This Phase III, randomized, open-label study was conducted at 3 sites in Thailand (WHO UTN: U1111-1233-9694; ClinicalTrials.gov: NCT04429295). The study protocol and 2 amendments were approved by the appropriate institutional review board, and the study was performed according to ethical principles derived from international guidelines, including the Declaration of Helsinki and the International Council for Harmonization Guidelines for Good Clinical Practice. An informed consent form was signed by each participant's parent(s) or legally acceptable representative(s) before enrolment. The study was conducted between June 2020 and May 2021.

Healthy 2-month-old, full-term infants (≥37 weeks of pregnancy) with a birth weight of ≥2.5 kg or medically stable preterm infants (27–36 weeks) were eligible for inclusion. All participants received Bacillus Calmette-Guérin vaccine ≥4 weeks before the study, HB vaccination ≤4 weeks after birth (except for 2 participants and not known for 1 participant), and could have received bOPV at any point during the study and influenza vaccine ≥2 weeks before or after any study vaccination. Exclusion criteria were: ongoing, recent (≤4 weeks prefirst vaccination), or planned participation in another clinical study; any vaccination in the 4 weeks prefirst study vaccination or planned receipt of any nonstudy vaccine in the 4 weeks before or after each study vaccination (other than OPV and influenza); previous vaccination against D, T, P, HB (except the HB birth dose), Hib, poliomyelitis (except OPV), rotavirus, or *Streptococcus pneumoniae*; immune globulins or blood products since birth; known or suspected immunodeficiency, immunosuppressive therapy since birth, or long-term corticosteroid therapy; history of blood dyscrasia, leukemia, lymphoma, neurologic disorder, intussusception, thrombocytopenia, or bleeding disorder; known personal or maternal history of HIV, HB surface antigen (HBsAg), hepatitis C seropositivity; history of D, T, P, poliovirus, HB, Hib, rotavirus, or pneumococcal infection; chronic or acute illness that could interfere with study conduct or evaluation of the objectives; febrile illness on the day of vaccination; receipt of antibiotics within 72 hours prefirst vaccination blood sample.

Infants were randomized (1:1 ratio) using interactive response technology with a permuted block method and stratified by the site to receive either DTwP-IPV-HB-PRP~T (2, 4 and 6 months of age) or separately administered DTwP-HB-PRP~T and bOPV (2, 4 and 6 months of age) and IPV (4 months of age). All participants received co-administered PCV13 (2, 4 and 6 months of age) and rotavirus vaccine (2 and 4 months of age).

The DTwP-IPV-HB-PRP~T and DTwP-HB-PRP~T vaccines were administered by intramuscular (IM) injection into the upper outer aspect of the left thigh, IPV was administered by IM injection into the upper outer aspect of the right thigh and bOPV was administered orally. The PCV13 (IM) and rotavirus (oral) vaccines were administered according to the manufacturer's instructions.

Study Vaccines

The hexavalent DTwP-IPV-HB-PRP~T vaccine [SHAN6, batches MHU001A19 (expiry February 2021) and MHU003A19 (expiry September 2021)] was manufactured by Sanofi Healthcare India Private Limited and supplied as a liquid, ready-to-inject, multidose (10 doses) vial presentation. Each 0.5 mL dose contained ≥30 IU D-toxoid (D antigen), ≥60 IU T-toxoid (T antigen), ≥4 IU whole-cell *B. pertussis* organisms (wP antigen), 10 µg rDNA HBsAg (HB antigen), 12 µg Hib purified capsular polysaccharide conjugated to 22–40 µg tetanus toxoid carrier protein [PRP~T (Hib) antigen], 29, 7 and 26 D antigen units of poliovirus type 1 (Mahoney strain), type 2 (MEF-1 strain) and type 3 (Saukett strain), respectively (IPV antigen), and 0.625 mg aluminum phosphate adjuvant (see Table, Supplemental Digital Content 2, http://links.lww.com/INF/F72). A discrepancy between the D antigen content versus IMOVAX Polio and other Sanofi IPV-containing vaccines was due to methodological changes and did not reflect a true difference in the D antigen content.

The pentavalent DTwP-HB-PRP~T vaccine (SHAN5, batch PLU017A19, expiry June 2021) was manufactured by Sanofi Healthcare India Private Limited and supplied as a liquid monodose vial presentation. Each 0.5 mL dose contained ≥30 IU D-toxoid (D antigen), ≥60 IU T-toxoid (T antigen), ≥4 IU whole-cell *B. pertussis* organisms (wP antigen), 10 µg rDNA HBsAg (HB antigen), 10 µg Hib purified capsular polysaccharide conjugated to 20–40 µg tetanus toxoid carrier protein [PRP~T (Hib) antigen], and 0.625 mg aluminum phosphate adjuvant (see Table, Supplemental Digital Content 3, http://links.lww.com/INF/F73).

The bOPV vaccine [batch T3B56 (expiry January 2021)] was manufactured by Sanofi and supplied as a liquid 20-dose vial presentation. Each 0.1 mL dose (2 drops) contained ≥6.0 log $CCID_{50}$ poliovirus type 1 LS c2ab and ≥5.8 log $CCID_{50}$ poliovirus type 3 Leon 12a1b (live attenuated Sabin strains) (see Tables, Supplemental Digital Content 3, http://links.lww.com/INF/F73 and Supplemental Digital Content 4 http://links.lww.com/INF/F74).

The IPV vaccine [IMOVAX Polio, batch T3B56 (expiry January 2021)] was manufactured by Sanofi and supplied as a liquid presentation. Each 0.5 mL dose contained 40, 8, and 32 D antigen units of poliovirus type 1 (Mahoney strain), type 2 (MEF-1 strain) and type 3 (Saukett strain), respectively (see Table, Supplemental Digital Content 5, http://links.lww.com/INF/F75).

The PCV13 vaccine [Prevnar13, batch AT2898 (expiry November 2021)] was manufactured by Pfizer and formulated as a suspension for injection. Each 0.5 mL dose contained 2.2 µg pneumococcal polysaccharide serotypes 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F and 4.4 µg pneumococcal polysaccharide serotype 6B. All serotypes were conjugated to CRM_{197} carrier protein adsorbed on aluminum phosphate, and each dose contained approximately 32 µg CRM_{197} and 0.125 mg aluminum (see Table, Supplemental Digital Content 6, http://links.lww.com/INF/F76).

Rotavirus vaccine [Rotarix, batch AROLC496AB (expiry February 2022)] was manufactured by GlaxoSmithKline and formulated as an oral solution in a single-dose container. Each 1.5 mL dose contained $\geq 10^{6.0}$ $CCID_{50}$ human rotavirus (live attenuated RIX4414 strain) (see Table, Supplemental Digital Content 7, http://links.lww.com/INF/F77).

Serology

Blood samples (approximately 3 mL) were collected from all participants prefirst vaccination and 28 days post-third vaccination for anti-D, anti-T, anti-pertussis, anti-Hib (PRP), anti-HB, anti-polio 1, anti-polio 2 and anti-polio 3 antibodies. Additionally, a randomized subset was used to evaluate the immune response to rotavirus vaccine (serum IgA anti-rotavirus antibody) and PCV13 [anti-pneumococcal immunoglobulin (IgG) for serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F].

All immunological assays were performed at Sanofi's Global Clinical Immunology laboratory (Swiftwater, PA), except for antirotavirus antibodies, which were assayed at the Laboratory for Specialized Clinical Studies at the Cincinnati Children's Hospital Medical Center.

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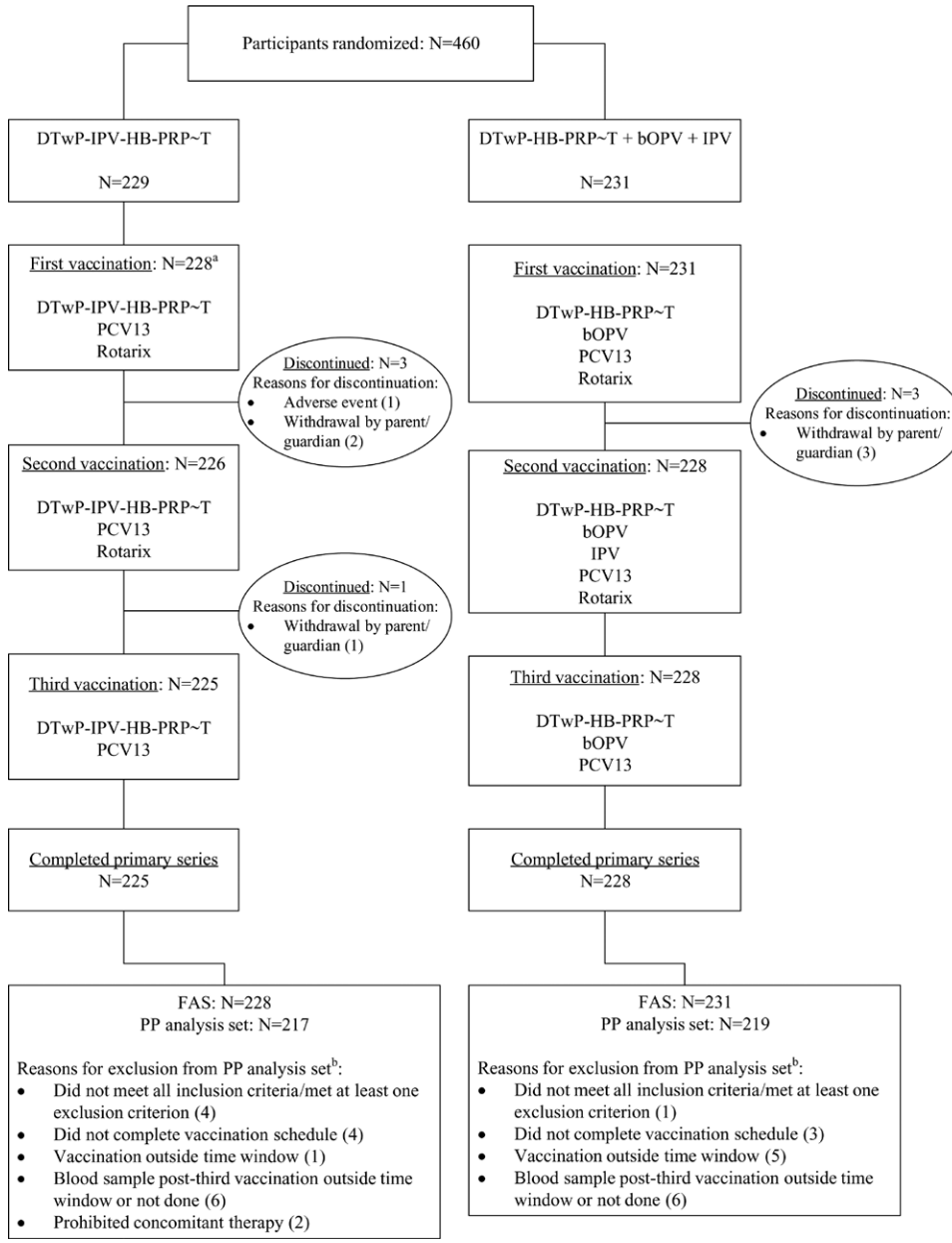


FIGURE 1. Disposition of study participants. [a]One participant was present but not vaccinated and no blood sample was performed due to withdrawal by the parent(s)/guardian. [b]A participant may have more than one reason for exclusion. FAS indicates full analysis set; PP, per protocol analysis set.

A validated Meso Scale Discovery Multiplexed Electro Chemoluminescence (DTP-ECL) immunoassay[6,7] was used to measure anti-D, anti-T, and anti-pertussis [anti-pertussis toxin (PT), anti-filamentous hemagglutinin (FHA), anti-pertactin (PRN), and anti-fimbriae 2/3 (FIM)] antibodies; anti-HBs antibodies were measured using a commercial immunometric technique (VITROS, Ortho Clinical Diagnostics, UK), anti-PRP antibody titers by an in-house radioimmunoassay, anti-poliovirus antibody titers by micro metabolic inhibition testing against wild type strains and anti-PCV13 antibodies by a Meso Scale Discovery Multiplexed Electro Chemoluminescence (IgG ECL) immunoassay. An in-house IgA enzyme immunoassay was used to measure anti-rotavirus antibodies.

Reactogenicity and Safety

Participants were observed for 30 minutes after each vaccination to assess immediate unsolicited adverse events (AEs). After leaving the study site, the duration and intensity [grade 1 (mild)–3 (severe)] of solicited injection site reactions (tenderness/erythema/swelling) and solicited systemic reactions (fever/vomiting/crying abnormal/drowsiness/appetite lost/irritability) reactions were recorded for 7 days after each vaccination using diary cards. All solicited reactions were automatically considered to be related to the vaccination. The axillary route was preferred for temperature measurement.

Unsolicited AEs were recorded using diary cards for 28 days after each vaccination. Unsolicited injection site AEs were

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automatically considered to be related to the vaccination and the investigators assessed unsolicited systemic AEs for causality and intensity.

Serious adverse events (SAEs) and AEs of special interest (AESIs) (anaphylaxis/hypersensitivity, convulsions, hypotonic hyporesponsive episodes, encephalopathy, extensive limb swelling) were collected throughout the study, and the investigators assessed their causality.

Statistical Analyses

The primary statistical objective was to demonstrate the noninferiority of the DTwP-IPV-HB-PRP~T vaccine to the DTwP-HB-PRP~T+bOPV+IPV vaccines based on immunogenicity. Secondary objectives were to describe the immune responses for DTwP-IPV-HB-PRP~T, DTwP-HB-PRP~T+bOPV+IPV, PCV13, and the rotavirus vaccine, and to describe the safety profile of DTwP-IPV-HB-PRP~T and DTwP-HB-PRP~T+bOPV+IPV. No statistical comparisons were performed for the secondary objectives.

Antibody thresholds and criteria used to define seroprotection (SP) and vaccine response (VR) rates are presented in Table 1 (noninferiority analysis) and Table 2. In addition to SP and VR rates, geometric mean titers (GMTs) (IPV) and geometric mean concentrations (GMCs) (D, T, HB, PRP, PT, FIM, PRN and FHA) and post-/pre-vaccination ratios (GMTRs or GMCRs) are presented. Anti-rotavirus and anti-PCV13 antibody GMCs and GMCRs are presented (Table 3) with seroconversion rates for anti-rotavirus IgA (≥4-fold rise) and SP rates for PCV13 serotypes (≥0.35 µg/mL). Data are shown with 95% confidence intervals (CIs), calculated using the exact binomial distribution (Clopper-Pearson)[8] for proportions and the normal approximation method for GMC and GMT ratios.

The noninferiority comparison was based on the 95% CI of the difference for SP rates (anti-D, anti-T, anti-PRP, anti-HB and anti-polio 1, 2, 3) and the ratio for adjusted GMCs (aGMCs) (anti-PT and anti-FIM). Adjusted GMCs were computed for anti-PT and anti-FIM using analysis of covariance to adjust for baseline disparities. For anti-D, anti-T, anti-PRP, anti-HB and anti-polio 1, 2, 3, noninferiority was concluded if the lower limit of the 2-sided CI for the difference was >−10%; for anti-PT and anti-FIM, noninferiority was concluded if the lower limit of the 2-sided 95% CI of the ratio was >0.5 for both antigens. Overall noninferiority was concluded if demonstrated separately for anti-D, anti-T, anti-PRP, anti-HB and anti-polio 1, 2, 3, and for the pertussis antigens (anti-PT and anti-FIM).

The planned sample size was 460 participants (230 participants per group), providing an overall power of 90% for the noninferiority analysis, assuming a 15% attrition rate. A subset of participants was randomized for the assessment of the rotavirus vaccine (115 participants per group) and PCV13 immunogenicity (the remaining 115 participants per group).

Immunogenicity analyses, including noninferiority testing, used the per-protocol (PP) population (participants with no protocol violation that could have interfered with the evaluation criteria, analyzed according to the vaccine received). Data from the full analysis set (those who received at least 1 vaccination, analyzed according to the randomization) supported the evaluation using the PP population. The safety evaluation used the safety analysis set (SafAS) (participants who received at least 1 vaccination).

The statistical analyses were performed under the responsibility of Sanofi's biostatistics group using SAS software, Version 9.4 (SAS Institute, Cary, NC).

RESULTS

Participants Studied

Overall, 460 participants were enrolled, with 229 randomized to DTwP-IPV-HB-PRP~T and 231 randomized to DTwP-HB-PRP~T+bOPV+IPV. A total of 459 participants received at least 1 vaccination (1 participant randomized to DTwP-IPV-HB-PRP~T was withdrawn by the parent(s)/guardian before administration of the first dose), and 225 participants (DTwP-IPV-HB-PRP~T) and 228 participants (DTwP-HB-PRP~T+bOPV+IPV) completed the study. Demographic characteristics were similar in each group, including maternal vaccination history, which was 65.1% and 70.6% (tetanus), 56.8% and 62.8% (diphtheria), 39.7% and 44.6% (pertussis) for the DTwP-IPV-HB-PRP~T and DTwP-HB-PRP~T+bOPV+IPV groups, respectively (see Table, Supplemental Digital Content 8, http://links.lww.com/INF/F78). Participant disposition, including reasons for discontinuation and exclusion from the PP population, is presented in Fig. 1.

Immunogenicity

Noninferiority was demonstrated for anti-D, anti-T, anti-PRP, anti-HB and anti-polio 1, 2, 3 (using SP rates) and anti-PT and anti-FIM (aGMCs) post-third vaccination, and overall noninferiority

TABLE 1. Noninferiority Analysis for DTwP-IPV-HB-PRP~T versus DTwP-HB-PRP~T+bOPV+IPV at 1 Month After the Three-dose Primary Vaccination Sseries (PP analysis set)

				DTwP-IPV-HB-PRP~T versus DTwP-HB-PRP~T+bOPV+IPV		
Antigen	Seroprotection Threshold*	DTwP-IPV-HB-PRP~T	DTwP-HB-PRP~T+bOPV+IPV	Difference or Ratio† (95% CI)	Delta	Noninferiority
Anti-D	≥0.01 IU/mL	100 (98.3;100)	100 (98.3;100)	0.00 (−1.74;1.72)	−10%	Yes
Anti-T	≥0.01 IU/mL	100 (98.3;100)	100 (98.3;100)	0.00 (−1.74;1.72)	−10%	Yes
Anti-PRP	≥0.15 µg/mL	100 (98.3;100)	99.5 (97.5;100)	0.46 (−1.32;2.54)	−10%	Yes
Anti-HBs	≥10 mIU/mL	98.6 (96.0;99.7)	98.6 (96.0;99.7)	−0.01 (−2.77;2.72)	−10%	Yes
Anti-polio 1	≥8 (1/dil)	100 (98.3;100)	100 (98.3;100)	0.00 (−1.74;1.72)	−10%	Yes
Anti-polio 2	≥8 (1/dil)	100 (98.3;100)	98.2 (95.4;99.5)	1.83 (−0.25;4.60)	−10%	Yes
Anti-polio 3	≥8 (1/dil)	100 (98.3;100)	100 (98.3;100)	0.00 (−1.76;1.72)	−10%	Yes
Anti-PT	aGMC	44.6 (37.3;53.3)	64.7 (54.1;77.3)	0.690 (0.536;0.888)	0.5	Yes
Anti-FIM	aGMC	937.3 (807.6;1087.7)	1150.4 (992.0;1334.2)	0.815 (0.660;1.01)	0.5	Yes

*Not applicable for pertussis antigens (aGMC presented for anti-PT and anti-FIM).

†Difference for anti-D, anti-T, anti-PRP, anti-HBs, anti-polio 1, 2, 3; ratio for anti-PT and anti-FIM.

Data are % (95% CI) (anti-D, anti-T, anti-HBs, anti-polio 1, 2, 3, and anti-PRP) or aGMC (95% CI) (anti-PT and anti-FIM).

For anti-D, anti-T, and anti-PRP, anti-HB, anti-polio 1, 2, 3, non-inferiority was concluded if the lower limit of the 2-sided CI for the difference was greater than 10%.

For anti-PT and anti-FIM, non-inferiority was concluded if the lower limit of the 2-sided 95% CI of the ratio was greater than 0.5.

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TABLE 2. Seroprotection Rates, Vaccine Response Rates, Geometric Mean Concentrations, and Geometric Mean Titers for DTwP-IPV-HB-PRP~T and DTwP-HB-PRP~T+bOPV+IPV Pre-first Vaccination and Post-third Vaccination (PP Analysis Set)

		DTwP-IPV-HB-PRP~T		DTwP-HB-PRP~T+bOPV+IPV	
Antigen	Variable	Pre-first Vaccination	Post-third Vaccination	Pre-first Vaccination	Post-third Vaccination
Anti-D	≥0.01 IU/mL	95.9 (92.3;98.1)	100 (98.3;100)	95.0 (91.2;97.5)	100 (98.3;100)
	≥0.1 IU/mL	55.8 (48.9;62.5)	99.1 (96.7;99.9)	50.2 (43.4;57.0)	100 (98.3;100)
	≥1.0 IU/mL	5.1 (2.6;8.9)	82.0 (76.3;86.9)	1.8 (0.5;4.6)	85.4 (80.0;89.8)
	GMC	0.104 (0.088;0.124)	1.86 (1.65;2.09)	0.093 (0.077;0.111)	2.09 (1.89;2.31)
	GMCR	NA	17.8 (14.0;22.7)	NA	22.6 (17.6;28.9)
Anti-T	≥0.01 IU/mL	100 (98.3;100)	100 (98.3;100)	100 (98.3;100)	100 (98.3;100)
	≥0.1 IU/mL	99.1 (96.7;99.9)	100 (98.3;100)	97.3 (94.1;99.0)	100 (98.3;100)
	≥1.0 IU/mL	66.4 (59.7;72.6)	96.3 (92.9;98.4)	63.0 (56.2;69.4)	97.3 (94.1;99.0)
	GMC	1.36 (1.19;1.56)	4.51 (4.09;4.97)	1.20 (1.03;1.40)	4.42 (4.02;4.86)
	GMCR	NA	3.30 (2.79;3.91)	NA	3.69 (3.07;4.43)
Anti-HBs	≥10 mIU/mL	26.4 (20.6;32.8)	98.6 (96.0;99.7)	25.3 (19.7;31.7)	98.6 (96.0;99.7)
	≥100 mIU/mL	9.7 (6.1;14.5)	93.5 (89.4;96.4)	6.5 (3.6;10.6)	96.8 (93.5;98.7)
	GMC	6.35 (5.18;7.79)	1694 (1349;2128)	5.99 (4.97;7.21)	2058 (1672;2532)
	GMCR	NA	271 (205;358)	NA	342 (266;440)
Anti-polio 1	≥8 (1/dil)	42.4 (35.7;49.3)	100 (98.3; 100)	49.3 (42.5;56.1)	100 (98.3; 100)
	GMT	6.47 (5.51;7.60)	2272 (1992;2592)	6.70 (5.85;7.66)	1516 (1320;1742)
	GMTR	NA	351 (279;442)	NA	226 (189;271)
Anti-polio 2	≥8 (1/dil)	57.5 (50.6;64.2)	100 (98.3; 100)	55.1 (48.2;61.8)	98.2 (95.4;99.5)
	GMT	9.10 (7.83;10.60)	2355 (2085;2660)	8.11 (6.99;9.41)	53.2 (47.3;59.8)
	GMTR	NA	256 (209;312)	NA	6.52 (5.34;7.96)
Anti-polio 3	≥8 (1/dil)	28.6 (22.7;35.1)	100 (98.3; 100)	33.9 (27.7;40.6)	100 (98.3; 100)
	GMT	5.30 (4.74;5.93)	3040 (2649;3488)	5.97 (5.35;6.67)	1407 (1242;1595)
	GMTR	NA	579 (484;692)	NA	236 (200;278)
Anti-PRP	≥0.15 µg/mL	30.1 (24.1;36.7)	100 (98.3; 100)	26.7 (21.0;33.1)	99.5 (97.5;100)
	≥1 µg/mL	3.7 (1.6;7.2)	96.3 (92.9;98.4)	1.8 (0.5;4.7)	94.5 (90.6;97.1)
	GMC	0.078 (0.064;0.094)	15.2 (13.0;17.8)	0.071 (0.060;0.083)	11.2 (9.50;13.1)
	GMCR	NA	196 (152;253)	NA	157 (125;197)
Anti-PT	≥2 EU/mL	66.8 (60.1;73.0)	96.8 (93.5;98.7)	69.9 (63.3;75.9)	97.3 (94.1;99.0)
	VR*	NA	63.6 (56.8;70.0)	NA	69.4 (62.8;75.4)
	≥4-fold rise	NA	54.8 (48.0;61.6)	NA	60.3 (53.5;66.8)
	GMC (EU/mL)	6.55 (5.22;8.21)	44.2 (34.8;56.1)	6.38 (5.12;7.94)	65.3 (51.6;82.6)
	GMCR	NA	6.75 (4.41;10.3)	NA	10.2 (6.79;15.4)
Anti-FIM	≥2 EU/mL	88.0 (82.9;92.0)	100 (98.3;100)	87.2 (82.1;91.3)	99.1 (96.7;99.9)
	VR*	NA	94.9 (91.1;97.4)	NA	95.4 (91.8;97.8)
	≥4-fold rise	NA	89.4 (84.5;93.2)	NA	93.2 (89.0;96.1)
	GMC (EU/mL)	11.4 (9.31;14.0)	928 (798;1080)	10.7 (8.77;13.0)	1161 (985;1369)
	GMCR	NA	81.2 (60.3;109)	NA	109 (81.4;146)
Anti-PRN	≥2 EU/mL	48.8 (42.0;55.7)	94.0 (90.0;96.8)	50.7 (43.9;57.5)	98.6 (96.0;99.7)
	VR*	NA	72.4 (65.9;78.2)	NA	81.3 (75.5;86.2)
	≥4-fold rise	NA	64.5 (57.8;70.9)	NA	72.6 (66.2;78.4)
	GMC (EU/mL)	3.38 (2.72;4.19)	21.1 (17.9;24.9)	3.88 (3.07; 4.89)	39.3 (34.1;45.2)
	GMCR	NA	6.26 (4.88;8.02)	NA	10.1 (7.82;13.1)
Anti-FHA	≥2 EU/mL	90.3 (85.6;93.9)	100 (98.3;100)	91.3 (86.8;94.7)	100 (98.3;100)
	VR*	NA	57.1 (50.3;63.8)	NA	66.7 (60.0;72.9)
	≥4-fold rise	NA	37.3 (30.9;44.1)	NA	48.9 (42.1;55.7)
	GMC (EU/mL)	20.5 (16.1;26.0)	42.9 (38.4;48.0)	24.8 (19.4;31.7)	93.2 (83.4;104)
	GMCR	NA	2.10 (1.56;2.82)	NA	3.75 (2.73;5.16)

*VR rate defined for anti-PT, anti-FIM, anti-PRN, and anti-FHA as: if pre-first vaccination concentration is <4 × lower limit of quantification (LLOQ) then post-third vaccination concentration ≥4 × LLOQ; if pre-first vaccination concentration ≥4 × LLOQ then post-third vaccination concentration ≥prevaccination concentration.

GMC indicates geometric mean concentration;GMCR, geometric mean concentration ratio; GMT, geometric mean titer; GMTR, geometric mean titer ratio; NA, not applicable; VR, vaccine response rate.

Data are % (95% CI) participants with titer or concentration above threshold, VR, 4-fold rise from pre-first vaccination, GMT, or GMC.

of DTwP-IPV-HB-PRP~T versus DTwP-HB-PRP~T+bOPV+IPV was concluded (Table 1).

Seroprotection rates, GMCs, and GMTs prefirst vaccination were generally similar in each group and were high for some antigens [eg, anti-D ≥0.01 IU/mL: 95.9% (DTwP-IPV-HB-PRP~T) and 95.0% (DTwP-HB-PRP~T+bOPV+IPV); anti-T ≥0.01 IU/mL: 100% and 100%; anti-FIM ≥2 EU/mL: 88.0% and 87.2%; and anti-FHA ≥2 EU/mL: 90.3% and 91.3%] (Table 2).

Post-third vaccination, all seroprotection rates were high in each group (for anti-D, anti-T, anti-HB, anti-polio 1, 2, and 3 and anti-PRP), and all VR rates were high in each group for pertussis antigens (anti-PT, anti-FIM, anti-PRN and anti-FHA) (Table 2). While there was a trend for lower postvaccination antibody levels in the DTwP-IPV-HB-PRP~T group, the 95% CIs overlapped for the GMC or GMT ratios for all antigens except polio. Anti-polio 1, 2 and 3 GMTs were higher for the DTwP-IPV-HB-PRP~T group than the DTwP-HB-PRP~T+bOPV+IPV group. This was particularly pronounced for anti-polio 2 [GMT ratio: 256 (DTwP-IPV-HB-PRP~T) versus 6.52 (DTwP-HB-PRP~T+bOPV+IPV)], which was administered on only one occasion in the DTwP-HB-PRP~T+bOPV+IPV group (IPV at 4 months of age).

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TABLE 3. Anti-rotavirus Vaccine Antibody and Anti-pneumococcal Vaccine (by serotype) Antibody Seroprotection Rate, Geometric Mean Concentration, and Geometric Mean Concentration Ratio by Serotype Pre-first Vaccination and Post-third Vaccination (PP analysis set)

		DTwP-IPV-HB-PRP~T*		DTwP-HB-PRP~T+bOPV+IPV*	
Antigen	Variable	Pre-first Vaccination	Post-third Vaccination†	Pre-first Vaccination	Post-third Vaccination†
Anti-rotavirus	≥4-fold rise	NA	83.3 (74.9;89.8)	NA	80.0 (71.1;87.2)
	GMC (EIA-U/mL)	3.88 (3.75;4.02)	79.1 (58.1;108)	4.00 (3.74;4.27)	70.5 (50.2;98.9)
	GMCR	NA	20.9 (15.4;28.5)	NA	17.3 (12.3;24.4)
Anti-PCV					
Serotype 1	≥0.35 µg/mL	17.0 (10.4;25.5)	100 (96.6;100)	19.8 (12.9;28.5)	99.1 (95.1;100)
	GMC	0.148 (0.128;0.172)	3.97 (3.42;4.61)	0.153 (0.127;0.184)	3.83 (3.27;4.47)
	GMCR	NA	26.7 (22.0;32.5)	NA	24.8 (19.2;32.1)
Serotype 3	≥0.35 µg/mL	9.4 (4.6;16.7)	94.3 (88.1;97.9)	3.6 (1.0;9.0)	94.6 (88.7;98.0)
	GMC	0.101 (0.090;0.114)	0.858 (0.765;0.964)	0.098 (0.089;0.109)	0.743 (0.668;0.827)
	GMCR	NA	8.46 (7.16;10.0)	NA	7.53 (6.44;8.80)
Serotype 4	≥0.35 µg/mL	5.7 (2.1;11.9)	100 (96.6;100)	5.4 (2.0;11.4)	100 (96.8;100)
	GMC	0.098 (0.087;0.110)	2.43 (2.16;2.73)	0.092 (0.083;0.102)	2.49 (2.24;2.77)
	GMCR	NA	24.8 (20.7;29.6)	NA	26.9 (23.1;31.3)
Serotype 5	≥0.35 µg/mL	14.2 (8.1;22.3)	98.1 (93.4;99.8)	11.7 (6.4;19.2)	100 (96.8;100)
	GMC	0.113 (0.098;0.130)	2.91 (2.52;3.36)	0.115 (0.100;0.133)	3.03 (2.65;3.46)
	GMCR	NA	25.8 (20.8;32.0)	NA	26.2 (21.5;31.8)
Serotype 6A	≥0.35 µg/mL	25.5 (17.5;34.9)	99.1 (94.9;100)	19.8 (12.9;28.5)	99.1 (95.1;100)
	GMC	0.169 (0.137;0.209)	5.21 (4.46;6.09)	0.170 (0.142;0.204)	5.69 (4.90;6.62)
	GMCR	NA	30.8 (23.0;41.1)	NA	33.1 (26.2;42.0)
Serotype 6B	≥0.35 µg/mL	17.9 (11.2;26.6)	90.6 (83.3;95.4)	18.0 (11.4;26.4)	95.5 (89.9;98.5)
	GMC	0.149 (0.124;0.179)	2.19 (1.73;2.76)	0.153 (0.129;0.182)	2.65 (2.16;3.25)
	GMCR	NA	14.7 (10.9;19.8)	NA	17.1 (12.7;23.0)
Serotype 7F	≥0.35 µg/mL	6.6 (2.7;13.1)	100 (96.6;100)	10.8 (5.7;18.1)	100 (96.8;100)
	GMC	0.126 (0.111;0.144)	4.12 (3.70;4.58)	0.128 (0.110;0.149)	3.96 (3.55;4.41)
	GMCR	NA	32.6 (27.7;38.4)	NA	30.5 (25.1;37.0)
Serotype 9V	≥0.35 µg/mL	15.1 (8.9;23.4)	100 (96.6;100)	16.2 (9.9;24.4)	99.1 (95.1;100)
	GMC	0.136 (0.117;0.159)	3.26 (2.86;3.71)	0.146 (0.125;0.171)	3.01 (2.63;3.44)
	GMCR	NA	23.9 (19.6;29.1)	NA	20.5 (16.4;25.7)
Serotype 14	≥0.35 µg/mL	80 (71.1;87.2)	100 (96.6;100)	80.0 (71.3;87.0)	100 (96.8;100)
	GMC	0.928 (0.715;1.20)	10.9 (9.32;12.9)	0.876 (0.685;1.12)	11.5 (9.64;13.7)
	GMCR	NA	11.8 (8.35;16.5)	NA	13.4 (9.50;18.9)
Serotype 18C	≥0.35 µg/mL	24.5 (16.7;33.8)	100 (96.6;100)	22.5 (15.1;31.4)	100 (96.8;100)
	GMC	0.185 (0.153;0.223)	2.96 (2.63;3.34)	0.173 (0.147;0.203)	3.00 (2.67;3.37)
	GMCR	NA	16.0 (12.7;20.2)	NA	17.2 (13.9;21.2)
Serotype 19A	≥0.35 µg/mL	29.2 (20.8;38.9)	100 (96.6;100)	35.1 (26.3;44.8)	100 (96.8;100)
	GMC	0.226 (0.186;0.274)	3.51 (3.04;4.05)	0.218 (0.179;0.265)	3.60 (3.14;4.14)
	GMCR	NA	15.5 (11.8;20.4)	NA	16.4 (12.6;21.2)
Serotype 19F	≥0.35 µg/mL	28.3 (20.0;37.9)	100 (96.6;100)	29.7 (21.4;39.1)	98.2 (93.7;99.8)
	GMC	0.197 (0.161;0.243)	5.73 (5.05;6.50)	0.187 (0.154;0.228)	5.73 (4.81;6.82)
	GMCR	NA	29.0 (22.1;38.1)	NA	30.3 (23.0;39.9)
Serotype 23F	≥0.35 µg/mL	26.4 (18.3;35.9)	98.1 (93.4;99.8)	27.0 (19.0;36.3)	98.2 (93.7;99.8)
	GMC	0.173 (0.142;0.210)	2.29 (1.91;2.74)	0.190 (0.154;0.234)	2.53 (2.17;2.96)
	GMCR	NA	13.2 (9.96;17.6)	NA	13.2 (10.0;17.4)

*For rotavirus, N=108 (DTwP-IPV-HB-PRP~T) and N=105 (DTwP-HB-PRP~T+bOPV+IPV); for PCV13, N=106 (DTwP-IPV-HB-PRP~T) and N=112 (DTwP-HB-PRP~T+bOPV+IPV).
†Post-third vaccination of DTwP-IPV-HB-PRP~T or DTwP-HB-PRP~T (post-second vaccination of Rotarix).
GMC, geometric mean concentration; GMCR, geometric mean concentration rat; NA, not applicable; PCV, pneumococcal conjugate vaccine.
Data are % (95% CI) participants with concentration above threshold or GMC.

For the rotavirus vaccine, the immune response postvaccination was comparable for co-administration with DTwP-IPV-HB-PRP~T or DTwP-HB-PRP~T+bOPV+IPV. The respective 4-fold rise was 83.3 and 80.0, and GMCR was 20.9 and 17.3 (Table 3).

For PCV13, GMCs were similar in each group for each serotype prefirst vaccination and post-third vaccination; GMCR and the percentage of participants with antipneumococcal titers ≥0.35 µg/mL post-third vaccination were similar in each group for all serotypes (Table 3).

Safety and Tolerability

No immediate AEs were reported. Most participants experienced at least one solicited injection site reaction or solicited systemic reaction, and the incidence was similar in each group [99.1% (DTwP-IPV-HB-PRP~T) and 98.3% (DTwP-HB-PRP~T+bOPV+IPV)] (Table 4). There was a trend for higher reactogenicity in the DTwP-IPV-HB-PRP~T group, in which all solicited reactions except drowsiness were more frequently reported than in the DTwP-HB-PRP~T+bOPV+IPV group (Table 4).

The incidence of participants with at least 1 unsolicited AE was similar in each group [19.3% (DTwP-IPV-HB-PRP~T) and 19.5% (DTwP-HB-PRP~T+bOPV+IPV)]. Few were considered by the investigators to be related to the study vaccines (1.3% and 1.7% of participants) (Table 4).

Overall, 35 participants [18 participants (DTwP-IPV-HB-PRP~T) and 17 participants ([DTwP-HB-PRP~T+bOPV+IPV)] experienced at least 1 SAE. Of these, 2 SAEs were reported as AESIs [febrile convulsion (DTwP-IPV-HB-PRP~T) and febrile seizure (DTwP-HB-PRP~T+bOPV+IPV)] and both were considered

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TABLE 4. Immediate, Solicited, Unsolicited, and Serious Adverse Events (SafAS)

	DTwP-IPV-HB-PRP~T			DTwP-HB-PRP~T+bOPV+IPV		
Participants With at Least One:	n/M	%	(95% CI)	n/M	%	(95% CI)
Immediate unsolicited AE*	0/228	0	(0.0;1.6)	0/231	0	(0.0;1.6)
Solicited reaction†	226/228	99.1	(96.9;99.9)	226/230	98.3	(95.6;99.5)
Solicited injection site reaction†	219/228	96.1	(92.6;98.2)	209/230	90.9	(86.4;94.3)
Tenderness	210/228	92.1	(87.8;95.3)	202/230	87.8	(82.9;91.8)
Erythema	128/228	56.1	(49.4;62.7)	100/230	43.5	(37.0;50.2)
Swelling	98/228	43.0	(36.5;49.7)	89/230	38.7	(32.4;45.3)
Solicited systemic reaction†	225/228	98.7	(96.2;99.7)	225/230	97.8	(95.0;99.3)
Fever	157/228	68.9	(62.4;74.8)	130/230	56.5	(49.8;63.0)
Vomiting	75/228	32.9	(26.8;39.4)	67/230	29.1	(23.3;35.5)
Crying abnormal	206/228	90.4	(85.8;93.9)	207/230	90.0	(85.4;93.6)
Drowsiness	183/228	80.3	(74.5;85.2)	189/230	82.2	(76.6;86.9)
Appetite lost	141/228	61.8	(55.2;68.2)	114/230	49.6	(42.9;56.2)
Irritability	196/228	86.0	(80.8;90.2)	193/230	83.9	(78.5;88.4)
Unsolicited AE‡	44/228	19.3	(14.4;25.0)	45/231	19.5	(14.6;25.2)
Unsolicited AR‡	3/228	1.3	(0.3;3.8)	4/231	1.7	(0.5;4.4)
AE leading to study discontinuation§	1/228	0.4	(0;2.4)	0/231	0	(0;1.6)
SAE§	18/228	7.9	(4.7;12.2)	17/231	7.4	(4.3;11.5)
AESI§	1/228	0.4	(0;2.4)	1/228	0.4	(0;2.4)
Death§	0/228	0	(0.0;1.6)	0/231	0	(0.0;1.6)

*Collected within 30 minutes after vaccination.

†Collected within 7 days after each vaccination.

‡Collected within 28 days after each vaccination.

§Collected throughout the study.

AE indicates adverse event; AESI, adverse event of special interest; AR, adverse reaction; M, number of participants with available data; n, number of participants; SAE, serious adverse event.

by the investigators to be related to the study vaccination. The AESI of febrile convulsion (DTwP-IPV-HB-PRP~T) occurred on day 1 postfirst injection and led to the withdrawal of the participant; the AESI of febrile seizure (DTwP-HB-PRP~T+bOPV+IPV) occurred on day 1 post-third vaccination, and the participant remained in the study. A further SAE (acute febrile illness on day 1 post-third vaccination of DTwP-IPV-HB-PRP~T) was considered by the investigators to be vaccine-related, and the participant remained in the study. No other SAEs were considered to be vaccine-related, and there were no deaths.

DISCUSSION

This study demonstrated the noninferiority of the fully liquid hexavalent DTwP-IPV-HB-PRP~T vaccine compared to separate administration of DTwP-HB-PRP~T+bOPV+IPV in Thailand, with rotavirus and PCV13 vaccines co-administered in each group. For the pertussis antigens, anti-PT and anti-FIM were included in the noninferiority analysis because they are the main drivers of pathogenicity and *Bordetella pertussis* agglutination,[9–13] and as reported previously.[5] Additionally, in a subset of participants, there was no difference in the response to the rotavirus or PCV13 vaccines when co-administered with DTwP-IPV-HB-PRP~T or DTwP-HB-PRP~T+bOPV+IPV.

The magnitude of the immune responses to DTwP-IPV-HB-PRP~T and DTwP-HB-PRP~T+bOPV+IPV was comparable to previous clinical studies conducted in India[3–5] and studies of similar pentavalent and hexavalent vaccines (containing aP rather than wP antigens) have not shown any reduction of the immune response when co-administered with rotavirus vaccines or PCV.[14–17] Furthermore, previous studies have shown the immune responses to rotavirus vaccine[15–17] and PCV[15,17,18] not to be affected by co-administration with DTaP-based vaccines, which is aligned with our findings.

Some antigens showed high SP and/or GMCs or GMTs before the first study vaccination, possibly reflecting prenatal maternal vaccination (eg, of D, T, and some pertussis antigens), although maternal vaccination rates were similar in each group and there was no difference in prevaccination rates between groups. For the polio antigens, postvaccination GMTs were higher for DTwP-IPV-HB-PRP~T than DTwP-HB-PRP~T+bOPV+IPV, particularly for anti-polio 2, but SP was high and similar in each group, and the differences in GMT were not considered to be clinically important. The difference for anti-polio 2 was considered to be due to the single administration of poliovirus type 2 (included in IPV but not bOPV) in the DTwP-HB-PRP~T+bOPV+IPV group. Differences between groups for other antigens were not considered to be of clinical consequence.

Regarding the safety profile, there was a trend for higher reactogenicity in the DTwP-IPV-HB-PRP~T group, although the incidence of unsolicited AEs was similar in each group and the study was not powered to detect meaningful differences in AE incidences. Vaccine-related events of febrile convulsion, febrile seizure, and acute febrile illness were reported but wP-containing vaccines have historically been associated with AEs of this type, which can be distressing but benign.[19,20] The WHO has concluded that these do not constitute a contraindication for the use of wP vaccines.[21]

Limitations of the study included the sample size of the study, which was based on the primary objective. The study was not powered to statistically compare secondary immunogenicity or safety parameters, and the ability to detect SAEs that occurred at a low frequency was limited.

Overall, this study confirmed the suitability for DTwP-IPV-HB-PRP~T primary series vaccination, in combination with rotavirus and PCV13 vaccines, with a strong immune response that was noninferior to licensed comparator vaccines and an acceptable safety profile.

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